Regulatory data sheet (RDS) - Pharmaceutical market (human and veterinary medicines)

GELUCIRE 44/14 (Code: 3051)

Last update: 28 March 2023

1. DEFINITION

DEFINITION
Lauroyl macrogol-32 glycerides EP / Lauroyl polyoxyl-32 glycerides NF / Lauroyl macrogolglycerides (32) ChP

DENOMINATION	
INCI NAME (PCPC - International Nomenclature of Cosmetic Ingredients)	Hydrogenated coconut oil PEG-32 esters
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	H5ZC52369M (Lauroyl PEG-32 Glycerides)
ADDITIONAL INFORMATION ON THE DENOMINATION	LAUROYL PEG-32 GLYCERIDES LAUROYL POLYOXYLGLYCERIDES

2. FIELD OF USE

FIELD OF USE This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Alcoholysis reaction between PEG-32 and hydrogenated Coconut oil mainly containing triglycerides of Lauric acid

COMPOSITION	
Well defined multi-constituent substance composed of mono, di- and triglycerides and PEG-32 mono- and di- esters of Lauric acid	
Full Labeling (Key letters system) **$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$**	
Hydrogenated coconut oil PEG-32 esters	100 %
Other constituents (i.e. incidental ingredients)	
None	

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4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	5253
Canada MF (Type III – Excipient)	1988-07
For submission of the DMF in another country, please contact your local Gattefossé representative	

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	**Conforms to monograph**
European Pharmacopeia Current edition	Lauroyl Macrogolglycerides
USP/ NF Current edition	Lauroyl Polyoxyglycerides
Chinese Pharmacopeia Current edition	Lauroyl Macrogolglycerides (32)

6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓		✓	✓ (5) (USFA 172.736)	✓ (8)	✓ (8) (40CFR180.910)

EUROPEAN REGULATION
(1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption)
(2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units)
(3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether)
(4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin
USA REGULATION
- Veterinary drugs
FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety
(5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR)
(6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database)
- Pesticides
(8)EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)

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7. SPECIFIC PHARMACEUTICAL REGULATIONS

CHINA BUNDLING REVIEW			
(Generic name) Chinese name	**Registration N°**	**Company**	**CFDA/CDE Reference (http://www.cde.org.cn/yfb.do?method=main)**
Lauroyl Macrogol-32 Glycerides 月桂酰聚氧乙烯(32)甘油酯	F20210000519	Gattefossé SAS	Raw material has not been approved yet for use in marketed formulations in Joint-Review with drug applications. However, the excipient dossier has already been submitted (Status I).
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	**Ingredient Name**	**Use**	**Restrictions**
Approved	Lauroyl macrogolglycerides or Lauroyl polyoxylglycerides	Not available as an Active Ingredient in any application. Available for use as an Excipient Ingredient in: Biologicals, Export Only, Prescription Medicines. Not available as an Equivalent Ingredient in any application.	Not authorized in listed medicines.
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	**NHPID name**	**Route of administration/ Restrictions**
Approved chemical name	Lauroyl polyoxylglycerides	Role Non-medicinal: Purposes: Solubilizing Agent
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE					
Study type/ duration	**Route**	**Species**	**Test article**	**Results/Conclusions**	**Reference**
Acute irritation (OECD 404)	Dermal	Rabbit	GELUCIRE® 44/14 No dilution	Non Irritant Primary irritation index: 0.2	(Gattefossé data, GAT-03631, 2003)
Acute irritation (OECD 405)	Ocular	Rabbit	GELUCIRE® 44/14 No dilution	Slightly Irritant Ocular irritation index (acute): 7.3 Ocular irritation index (48h): 0.7	(Gattefossé data, GAT-03632, 2003)
Acute toxicity (OECD 401)	Oral	Rat	GELUCIRE® 44/14 pure	$LD50_{(oral)}$ > 2004 mg/kg	(Gattefossé data, GAT-97334, 1997)

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SAFETY STUDIES PERFORMED BY GATTEFOSSE					
Study type/ duration	**Route**	**Species**	**Test article**	**Results/Conclusions**	**Reference**
Subchronic 7-day toxicity study (DRf)	Oral (Gavage)	Rat	GELUCIRE® 44/14 Vehicle: none Control: comparable regimen Dose levels: 0, 600, 1500 and 2400 mg/kg/day	$MTD_{(oral)}$ > 2400 mg/kg/day $NOEL_{(oral)}$: 2400 mg/kg/day	(Gattefossé data, GAT- WIL261007, 2003)
Subchronic 28-day toxicity study	Oral (Gavage)	Rat	GELUCIRE® 44/14 Vehicle: none Control: comparable regimen Dose levels: 0, 600, 1500 and 2400 mg/kg/day	$NOEL_{(oral)}$ > 2400 mg/kg/day	(Gattefossé data, GAT- WIL261008, 2003)
Subchronic 13-week toxicity study	Oral (Capsule)	Dog	GELUCIRE® 44/14 Vehicle: none Control: comparable regimen Dose levels: 0, 400, 1000 and 2500 mg/kg/day	$NOAEL_{(oral)}$ > 2500 mg/kg/day	(Gattefossé data, GAT- WIL 261002, 1995)
Reprotoxicity study Embryo-fetal development (Segment II)	Oral (Gavage)	Rat	GELUCIRE® 44/14 Vehicle: none Control: comparable regimen Dose levels: 0, 600, 1500 and 2400 mg/kg/day	$NOAEL_{Matern}$: 1500 mg/kg/day $NOAEL_{Embryo/fetal}$: 2400 mg/kg/day	(Gattefossé data, GAT- WIL 261011, 2004)
Ames test (With and without activation) (OECD 471)	In vitro	S. typhimurium	GELUCIRE® 44/14	Negative. Not mutagenic.	(Gattefossé data, GAT- Pharmak PH301-GAF-001-95, 1996)
Mouse Lymphoma Assay	*In vitro*	Mouse lymphoma cell	GELUCIRE® 44/14	Negative.	Gattefossé data, GAT-Pharmak PH313-GAF-001-95, 1996)
Mammalian Erythrocyte Micronucleus Test	IP	Mouse	GELUCIRE® 44/14	Negative. Not clastogenic.	(Gattefossé data, Pharmak PH309-GAF-001-95, 1996)
OECD : Organisation for Economic Co-operation and development - JORF: Journal official République Française (Official Journal French Republic)					
Summary of files available upon request, please contact your Gattefossé local representative. Additional literature review is available for the different endpoints, please consult the document "Tox and safety overview" for a complete review on regulatory, tox and safety aspects.					

9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI NAME	**CAS #**	**EC #**	**Chemical name**
Hydrogenated coconut oil PEG-32 esters	57107-95-6 (or 9004-81-3 + 27638-00-2)	Not applicable (or Not applicable + 248-586-9)	Poly(oxy-1,2-ethanediyl), α,α',α''-1,2,3-propanetriyltris[ω-hydroxy-, dodecanoate (Or Poly(oxy-1,2-ethanediyl), α -(1-oxododecyl)- ω -hydroxy- + Dodecanoic diester with 1,2,3-propanetriol:
EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list. EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5. Substances without EC numbers were assigned "list numbers" in order to facilitate REACh submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.			

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NATIONAL INVENTORIES					
INCI NAME	**U.S.A. (TSCA)**	**Canada (DSL/NDSL)**	**Australia (AICS)**	**China (IECSC)**	**Other inventories**
Hydrogenated coconut oil PEG-32 esters	Listed	NDSL listed	Listed	Listed	Poly(oxy-1,2-ethanediyl), α,α′,α′′-1,2,3-propanetriyltris[ω-hydroxy-, dodecanoate: IECSC, NDSL, REACH, TCSI, TSCA, VNECI / Poly(oxy-1,2-ethanediyl), α,α',α''-1,2,3-propanetriyltris[ω-hydroxy-, dodecanoate: AIIC (Or) Poly(oxy-1,2-ethanediyl), α-(1-oxododecyl)-ω-hydroxy-: AIIC, DSL, ENCS, NZIoC, PICCS, REACH, TCSI, TSCA, VNECI / Poly(oxy-1,2-ethanediyl), α-(dodecanoyl)-ω-hydroxy-: IECSC / α-(1-Oxododecyl)-ω-hydroxypoly(oxy-1,2-ethanediyl): AREC, ECL + Dodecanoic acid, diester with 1,2,3-propanetriol: AIIC, DSL, ENCS, IECSC, NZIoC, PICCS, TCSI, TSCA / lauric acid, diester with glycerol: EINECS, REACH / Dodecanoic acid diester with 1,2,3-propanetriol: AREC, ECL / Diester of dodecanoic acid and 1,2,3-propanetriol: VNECI

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics… are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.

Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities.
Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration.
Chemicals not listed on the AIIC must be assessed before importation.

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	**Status per substance**
Hydrogenated coconut oil PEG-32 esters	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACh regulation.

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10. HISTORY OF USE

10.1. Pharmaceutical applications

10.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	**Route**	**Dosage form**	**Maximum Potency per unit dose**	**Maximum Daily Exposure (MDE)**
LAUROYL PEG-32 GLYCERIDES (UNII: H5ZC52369M)	ORAL	Capsule		7200mg
	ORAL	Tablet	0.15mg	
LAUROYL POLYOXYLGLYCERIDES (UNII: NA)	ORAL	Capsule	218 mg	
	ORAL	Tablet, film coated	0.15 mg	
	ORAL	Tablet	0.15 mg	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.2. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.
For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.’

The similarities may be based on the following:
- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	**Route**	**Dosage form**	**Maximum Potency per unit dose**	**Maximum Daily Exposure (MDE)**
STEAROYL POLYOXYLGLYCERIDES (UNII: NA)	ORAL	Capsule	260 mg	
	ORAL	Tablet	23.34 mg	
STEAROYL POLYOXYL-	ORAL	Capsule		164 mg

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FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	**Route**	**Dosage form**	**Maximum Potency per unit dose**	**Maximum Daily Exposure (MDE)**
32 GLYCERIDES (UNII: G6EP177239)	ORAL	Capsule, Extended Release		8 mg
	ORAL	Tablet		3 mg
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.3. FDA IIG references for larger family of chemicals

The following reference covers multiconstituant substances containing LAUROYL PEG-32 GLYCERIDES.

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	**Route**	**Dosage form**	**Maximum Potency per unit dose**	**Maximum Daily Exposure (MDE)**
*PEG VEGETABLE OIL** *(UNII: Not available)*	*IM – SC*	*Injection*	*7 %*	
	IV(INFUSION)	*Injection*	*5 %*	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

**removed from IIG in July 2019 due to new data standard. The reference was not removed from IIG due to safety concerns.*

10.1.4. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENT (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)			
Denomination	**Functional Category**	**Regulatory status**	**Safety**
Polyoxylglycerides	Emulsifying agent; modified-release agent; nonionic surfactant; penetration enhancer; solubilizing agent.	Lauroyl polyoxylglycerides and stearoyl polyoxylglycerides are approved as food additives in the US. Included in the FDA Inactive Ingredients Database (oral route: capsules, tablets, solutions; topical route: emulsions, creams, lotions; vaginal route: emulsions, creams). Oleyl polyoxylglycerides are included in a topical cream formulation licensed in the UK.	Polyoxylglycerides are used in oral and topical pharmaceutical formulations, and also in cosmetics and food products. They are generally regarded as relatively nonirritant and nontoxic materials.

10.2. Food/Nutraceutical/Dietary supplement applications

Polyoxylglycerides (including Lauroyl Polyoxylglycerides) have a long history of use in USA as food additive for use in nutraceutical products and dietary supplements.

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Country	Reference	Registration number
U.S.A.	USFA (US Food additives) Published in Federal register Vol. 71, No.48 (March 13, 2006)	21 CFR § 172.736 Food Additives Permitted For Direct Addition to Food for Human Consumption: Glycerides and Polyglycides of hydrogenated vegetable oils

Glycerides and Polyglycides of hydrogenated is authorised for use as food additive in nutraceutical products and dietary supplements.

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of GELUCIRE® 44/14 for use in nutraceutical products and dietary supplements remain the own responsibility of the company marketing the finished product.

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